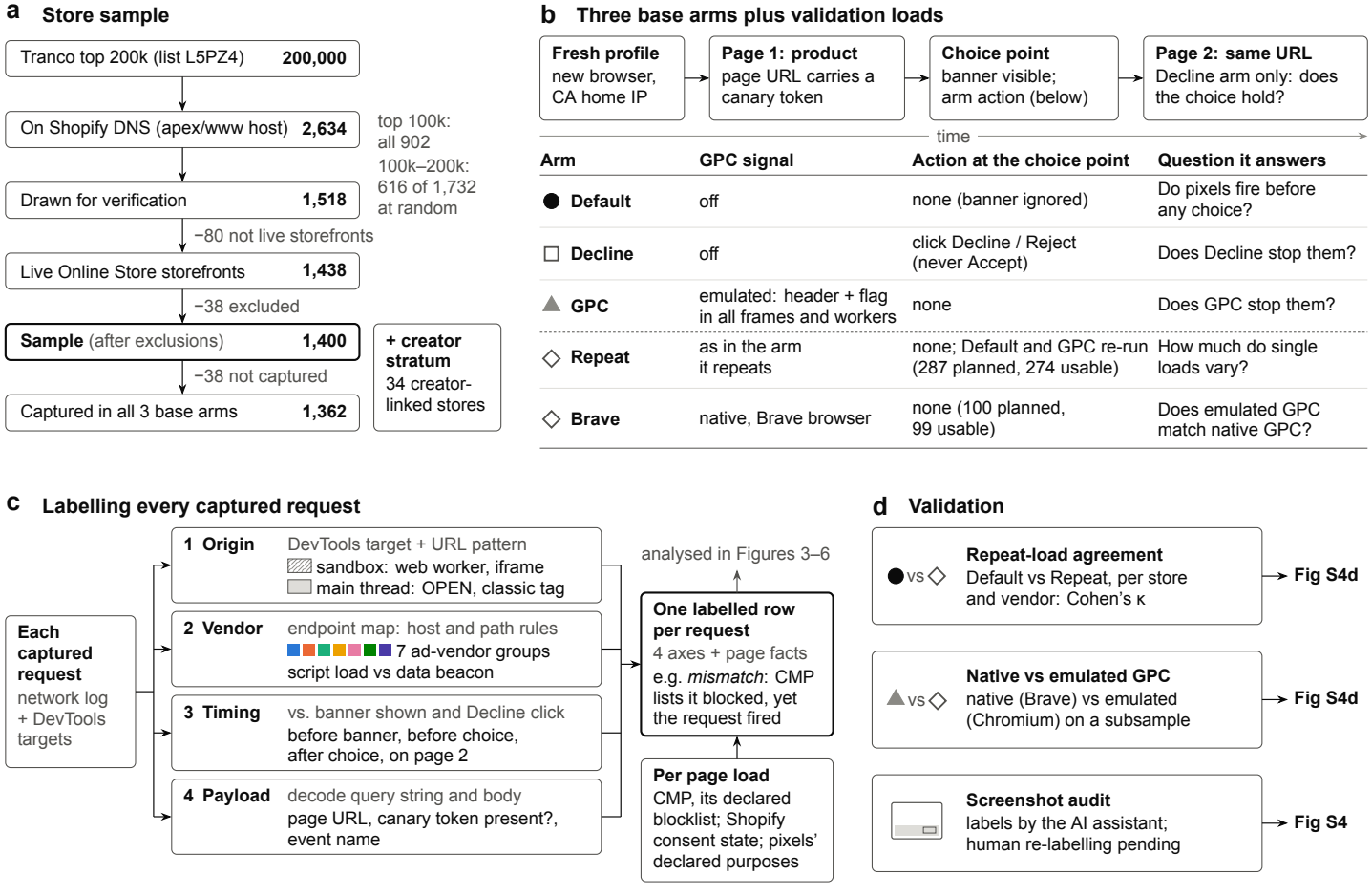

a
Store sample
Tranco top 200k (list L5PZ4) 200,000
On Shopify DNS (apex/www host) 2,634
top 100k: all 902
100k–200k: 616 of 1,732 at random
Drawn for verification 1,518
−80 not live storefronts
Live Online Store storefronts 1,438
−38 excluded
Sample (after exclusions) 1,400
−38 not captured
Captured in all 3 base arms 1,362
+ creator stratum
34 creator-linked stores
b
Three base arms plus validation loads
Fresh profile
new browser, CA home IP
Page 1: product
page URL carries a canary token
Choice point
banner visible; arm action (below)
Page 2: same URL
Decline arm only: does the choice hold?
time
Arm
GPC signal
Action at the choice point
Question it answers
Default
off
none (banner ignored)
Do pixels fire before any choice?
Decline
off
click Decline / Reject (never Accept)
Does Decline stop them?
GPC
emulated: header + flag in all frames and workers
none
Does GPC stop them?
Repeat
as in the arm it repeats
none; Default and GPC re-run (287 planned, 274 usable)
How much do single loads vary?
Brave
native, Brave browser
none (100 planned, 99 usable)
Does emulated GPC match native GPC?
c
Labelling every captured request
Each captured request
network log + DevTools targets
1 Origin
DevTools target + URL pattern
sandbox: web worker, iframe
main thread: OPEN, classic tag
2 Vendor
endpoint map: host and path rules
7 ad-vendor groups
script load vs data beacon
3 Timing
vs. banner shown and Decline click
before banner, before choice, after choice, on page 2
4 Payload
decode query string and body
page URL, canary token present?, event name
analysed in Figures 3–6
One labelled row per request
4 axes + page facts
e.g. mismatch: CMP lists it blocked, yet the request fired
Per page load
CMP, its declared blocklist; Shopify consent state; pixels' declared purposes
d
Validation
vs
Repeat-load agreement
Default vs Repeat, per store and vendor: Cohen's κ
Fig S4d
vs
Native vs emulated GPC
native (Brave) vs emulated (Chromium) on a subsample
Fig S4d
Screenshot audit
labels by the AI assistant; human re-labelling pending
Fig S4